PARTHIBAN M

AI Developer — Full Stack Developer — Automation Enthusiast

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Summary

- Results-driven AI and Full Stack Developer with strong expertise in building intelligent automation systems and real-world applications. Experienced in developing deep learning models, OCR pipelines, and scalable web solutions with a focus on performance, accuracy, and usability. Proven ability to design end-to-end systems from problem identification to deployment, with a keen interest in solving practical industry challenges using AI.

EDUCATION

- **B.Sc Computer Science** **(2023-2026)**
  GOVERNMENT ARTS COLLAGE COLLEGE
  (AUTONOMOUS) SALEM-636007
  CGPA: 7 / 10

SKILLS

- **Programming Languages**
  Python, JavaScript,PHP
- **Web Development**
  HTML, CSS, JavaScript, React, Node.js
- **AI & Automation**
  OCR, Chatbots, OpenAI APIs, Automation Scripts
- **Databases**
  SQLite, MySQL
- **Tools**
  Git, GitHub, VS Code, Google Colab

PROJECTS

- **DeepFingerprint: CNN-Based Latent Fingerprint Reconstruction & Inpainting**
  Developed a deep learning-based image reconstruction system using Convolutional Autoencoders (TensorFlow/Keras) to restore damaged or partial fingerprint images. Designed a synthetic corruption pipeline with geometric masking to simulate real-world degradation. Implemented a hybrid loss function (MSE + SSIM) to preserve ridge structure and texture. Enhanced output quality using CLAHE, denoising, and sharpening techniques to produce AFIS-compatible results.
  Tech: Python, TensorFlow, Keras, OpenCV, NumPy, Scikit-Image
- **Hybrid Multi-Engine OCR & Document Digitization Pipeline**
  Built an intelligent document processing pipeline to convert scanned images and PDFs into structured text. Implemented a fallback architecture that combines direct PDF parsing with OCR-based extraction. Integrated OpenCV preprocessing and leveraged Tesseract & EasyOCR for robust multi-engine recognition across noisy and unstructured inputs.
  Tech: Python, OpenCV, Tesseract OCR, EasyOCR, PyPDF2, PDF2Image

- **Medical Shop Management Web Application**
  Built a database-driven web application to manage medicine inventory with add/view operations and persistent storage using SQLite. Focused on efficient data handling and user-friendly interface.
  Tech: HTML, CSS, JavaScript, MySQL

- **CLOAK CV-VISION — Real-Time Invisible Cloak**
  Developed a real-time computer vision application using Python and OpenCV to create an Invisible Cloak effect through live webcam processing. Implemented background capture, HSV-based red color segmentation, mask refinement, morphological filtering, and background replacement with alpha blending for smooth output.
  Tech: Python, OpenCV, NumPy, Computer Vision, HSV Segmentation

Significant Roles

- **Freelance Developer** **2024 – Present**
  Remote
  Worked on real-time client projects including AI chatbots, automation tools, and web applications.

Hobbies & Interests

- Building experimental AI projects, exploring emerging technologies, solving real-world problems through automation, and actively learning new tools in software development.

Declaration

- I hereby declare that the above mentioned information is correct to the best of my knowledge.